

The Weakness of a Single Decision Tree fragility

A single decision tree, while easy to understand, has a significant weakness: it is **highly sensitive** to small changes in the training data.

- A decision tree is **not robust**.
 - **Example:** Changing just **one** training example out of ten can cause the algorithm to pick a completely different feature for the root node split (e.g., "Whiskers" instead of "Ear Shape").
 - This single change at the root leads to a **completely different tree structure**, which can result in different predictions.
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The Solution: Tree Ensembles

Instead of relying on one "unstable" tree, a more robust approach is to build **many** different decision trees and have them work together.

- A **tree ensemble** is a collection or "ensemble" of multiple decision trees.
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How Ensembles Make Predictions: Voting

The core idea of a tree ensemble is to let all the trees vote on the final prediction.

1. A new test example is fed into **every single tree** in the ensemble.
2. Each tree makes its own independent prediction.
3. The final prediction is the class that receives the **majority of the votes**.

Example:

You want to classify a new animal. You run it through your ensemble of 3 trees:

- **Tree 1 predicts:** "Cat"
- **Tree 2 predicts:** "Not Cat"
- **Tree 3 predicts:** "Cat"

The **majority vote** is 2-to-1 for "Cat." Therefore, the ensemble's final prediction is **"Cat"**.

Why Ensembles Work

- **Robustness:** By averaging the "opinions" of many trees, the overall algorithm becomes much **less sensitive** to the individual flaws or quirks of any one tree.
- The final model is more stable and often much more accurate than any single decision tree.

The key challenge, which will be addressed next, is how to build *many different and plausible* trees from the same dataset. This is achieved using a technique called **sampling with replacement**.